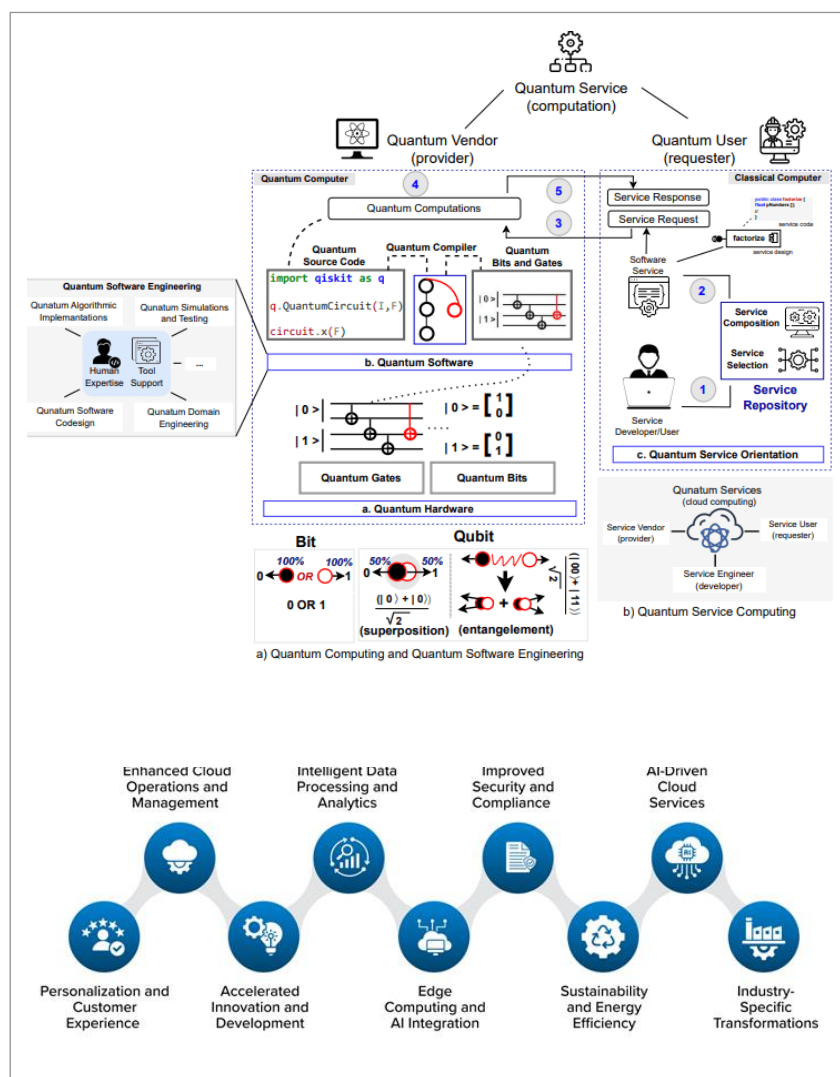




and quantum computing services are being combined into a single environment.

## LLM infrastructure at scale: Open source AI cloud stack

Large language models have quickly become the core of many modern applications, ranging from intelligent chatbots and coding assistants to research and enterprise knowledge systems. To deploy large language models at scale, however, it takes significantly more than just running a simple machine learning script — it requires the development of infrastructure that can handle enormous computational demands while remaining efficient and reliable.

The core of any AI-based cloud infrastructure is container orchestration. Container orchestration platforms, such as Kubernetes, enable developers to efficiently handle distributed applications across a series of machines.

GPUs are essential in modern AI systems. Technologies such as NVIDIA CUDA enable the use of thousands of cores in parallel computations, which are necessary to reduce the time taken in machine learning model training and inference. These GPUs can be dynamically assigned to various applications based on their needs using container orchestration platforms. In

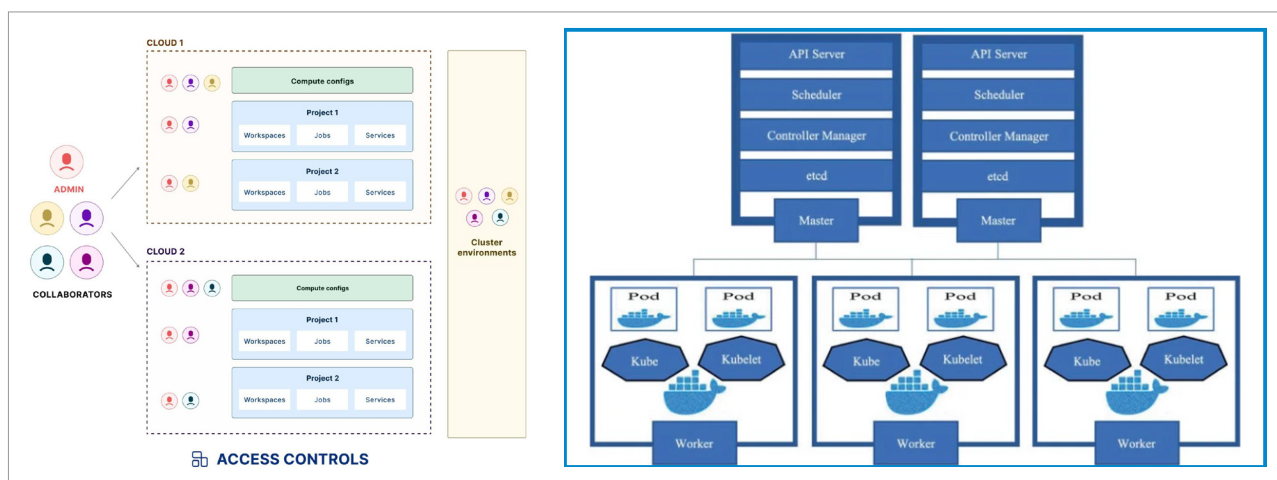


Figure 2: Container orchestration across series